

## SECTION B

7 (**short**) A geostationary satellite orbits the Earth directly above the equator, at the same rotational speed as the Earth's rotation. This means that it remains fixed over a specific point on the Earth's surface.

(a) Find the altitude of the satellite above the Earth's surface. [6]

(b) Calculate the work done on the satellite to launch it into orbit from Earth. Assume that the take-off site is equatorial and that the initial kinetic energy due to Earth's rotation may be neglected. [4]

|                        |     |                                                      |
|------------------------|-----|------------------------------------------------------|
| Gravitational constant | $G$ | $6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ |
| Radius of the Earth    | $R$ | 6380 km                                              |
| Mass of the Earth      | $M$ | $5.97 \times 10^{24} \text{ kg}$                     |
| Mass of the satellite  | $m$ | 1000 kg                                              |